



Laminar burning velocity of acetic acid + air flames



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ABSTRACT

Laminar burning velocities of acetic acid + air flames at 1 atm and initial gas temperatures of 338 K, 348 K, and 358 K were determined using the heat flux method. Measurements were performed in non-stretched flames, stabilized on a perforated plate burner at adiabatic conditions. Due to experimental problems related to the corrosiveness of acetic acid towards the burner material, the uncertainty of the burning velocities was relatively high up to ± 2 cm/s. Seventy reactions pertinent to acetic acid and ketene have been reviewed and detailed reaction mechanism for acetic acid combustion was developed. The model over-predicts measured burning velocities by about 3 cm/s. The mechanism was also tested comparing with flame structure of the low-pressure flame of acetic acid (Leplat and Vandooren, 2012). Good agreement with the concentration profiles of major products was found, however several minor intermediates were over- or under-predicted by the model. To elucidate reactions responsible for the differences observed, the sensitivity analysis was performed. It was found that the calculated burning velocities are insensitive to the reactions of acetic acid and mostly governed by C1 chemistry typical for all hydrocarbons and by reactions of ketene. Possible modifications of the rate constants within the evaluated uncertainty factors were discussed.

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1. Introduction

Carboxylic acids are abundant species in the environment that originate from both anthropogenic and biogenic sources. Their presence in soil, rain water and in the atmosphere has been well documented [1]. Acetic acid, $\text{CH}_3\text{CO}_2\text{H}$, is one of the most common carboxylic acids with high concentrations registered in urban areas. It is believed that the increased levels of acetic as well as other carboxylic acids are caused by traffic, a claim supported by many studies. For instance, Kawamura et al. [2] investigated emissions of gasoline engine and suggested motor exhaust to be the important source for organic acids including $\text{CH}_3\text{CO}_2\text{H}$. The same conclusion was reached by Talbot et al. [3], who studied direct emissions from vehicles and bio-mass burning. It was seen that both sources generated large quantities of $\text{CH}_3\text{CO}_2\text{H}$. The presence of carboxylic acids in the atmosphere in an urban area of São Paulo was measured by Souza et al. [4]. The results indicated that the acids showed both diurnal and nocturnal variations, with increased concentrations during night time. The authors suggested that this could be caused by photochemical reactions and/or emissions directly from fossil fuel combustion. Zervas et al.

investigated emissions of a spark ignition [5] and compression ignition [6] engines. Several fuels were investigated and the results indicated that the emissions of $\text{CH}_3\text{CO}_2\text{H}$ were increased when the equivalence ratio was decreased. Addition of toluene was seen to enhance the emission of acetic acid [5]. Similar results were found in the study of the compression engine [6]. Although aldehydes were sometimes considered to be precursors of acids, it is noteworthy that Zervas et al. [6] measured a 6.5 times larger formation of organic acids than aldehydes.

In order to understand the formation of organic acids during combustion, Zervas [7] experimentally measured species profiles of burner stabilized propane, isooctane and toluene/isooctane flames using chromatography with a flame ionization detector. Maximum concentration of $\text{CH}_3\text{CO}_2\text{H}$ was found at lean conditions in all flames. This is in agreement with the engines results [5,6]. A clear relation between the formation of C2 radicals and the production of $\text{CH}_3\text{CO}_2\text{H}$ was observed, as C2 radical species are considered to be major precursors of $\text{CH}_3\text{CO}_2\text{H}$. To further investigate the formation mechanism of acetic and other acids, Battin-Leclerc et al. [8] developed a model and simulated the experimental results from the propane flames by Zervas [7]. While the mechanism was in acceptable agreement with these scarce experimental data, it failed at reproducing the increasing emissions of $\text{CH}_3\text{CO}_2\text{H}$ experimentally observed at leaner conditions. The mechanism presumed that the formation of acetic acid is mainly driven by the $\text{CH}_3\text{CO} + \text{OH}$ reaction.

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Further validation and improvement of the model developed by Battin-Leclerc et al. [8] was hampered by the lack of experimental studies of acetic acid combustion. Essentially, only two reports were available: on investigation of the flammability limits of acetic acid [9], and on its oxidation in a flow reactor [10]; a vast majority of kinetic studies outlined in the following was focused on $\text{CH}_3\text{CO}_2\text{H}$ thermal decomposition or reactions at atmospheric conditions. Most recently, acetic acid flames stabilized at 50 mbar were investigated by Leplat and Vandooren [11]. Species profiles were obtained using molecular beam mass spectrometer and a kinetic model was created to simulate the experimental results. The detailed mechanism of Leplat and Vandooren was based on the ethanol mechanism developed by the same group [12]. Additional acetic acid reactions important for the combustion were added and the choice of their rate constants was outlined. While the mechanism was mainly constructed using the rate constants suggested in the literature, some modifications and assumption were made in order to improve the model behavior.

The chemistry of acetic acid is of importance in combustion of renewable oxygenated fuels, namely ethyl esters [13,14]. These esters may decompose directly forming $\text{CH}_3\text{CO}_2\text{H}$, and therefore Gasnot et al. [13] introduced several reaction of $\text{CH}_3\text{CO}_2\text{H} + \text{radicals}$ assuming H atom abstraction from methyl group forming $\text{CH}_2\text{CO}_2\text{H}$ only and assigning generic rate constants suggested by Warnatz [15] for reactions with primary H atoms of hydrocarbons. Leplat and Vandooren [11] introduced several reactions of $\text{CH}_3\text{CO}_2\text{H} + \text{radicals}$ for H atom abstraction from acidic group forming CH_3CO_2 and assumed these rate constants similar to H atom abstraction from alcohol group. Ren et al. [14] constructed a model for C3–C5 ethyl esters decomposition and adopted several reactions of $\text{CH}_3\text{CO}_2\text{H} + \text{radicals}$ from Leplat and Vandooren [11] without modification. Meanwhile theoretical studies of H atom abstraction reactions from acetic acid [16] and other relevant species have been published that can certainly help improving the acetic acid kinetic sub-mechanism.

The goal of the present study was therefore twofold: (a) to determine the laminar burning velocities, S_L , of premixed acetic acid + air flames at atmospheric pressure, and (b) to extend a recently updated detailed kinetic mechanism for small oxygenated fuels [17] by reactions of acetic acid and validate it. Moreover, the present mechanism was tested using experimental data from the low-pressure flames [11].

2. Experimental details

The heat flux method was used to determine the laminar burning velocities of premixed acetic acid + air flames at atmospheric pressure. The acetic acid was provided by Alfa Aesar and had a purity of 99.7%. Basic principles behind the heat flux method have been described in detail in many previous publications, e.g. [18–24]; therefore only specific features of the present measurements will be discussed. Schematic of the experimental setup employed in the present study was originally depicted in [20] with the latest modifications shown and described by Christensen et al. [21]. Design of the heat flux burner was essentially the same as introduced by Bosschaart and de Goey [18] with cross-section and technical details of the burner head and of the burner plate presented recently in [22,23].

All heat flux burners employed so far in our and other laboratories are made of brass; hence the main challenge of measuring the burning velocity was the corrosiveness of acetic acid towards the burner material. Although acetic acid is relatively weak, it is well known that it may react with brass [25], copper [26], bronze and similar metals producing patina on the surface. Thus it was expected that the burner exposed to acetic acid could be spoiled. For this reason the burner with the worst characteristics (signifi-

cant asymmetry in the temperature profile) designated as burner “E2” by Alekseev et al. [19] was employed. This burner previously has been used in a “surface area” experiment. As described in [19], some of the holes of the burner plate close to its edge were covered with thermal paste in order to investigate how a decrease of the surface area affects the measured laminar burning velocity of methane. It should be emphasized though that the holes were cleaned from the thermal paste prior to the start of the present experiments. Nevertheless, the burning velocities of methane measured prior to the acetic acid experiments were found significantly lower (by 4.5 cm/s at stoichiometric conditions) than the results of the reference burner (burner “TS” [19]) as illustrated in Supplemental material, Fig. S1. The reference burner was evaluated to have an uncertainty below ± 0.5 cm/s [19]. Two factors may contribute to this difference. First, the burner “E2”, even before any modification, showed burning velocities by 1–2 cm/s lower than the reference burner [19] and therefore was not used in our published experiments. Second, although the burner holes cleaned from the thermal paste were visually open, some residual material may reduce the effective hole size and thus modify the overall effective surface area of the burner plate.

To circumvent the limitations described above the following procedure was adopted: before each measurement series with acetic acid the burning velocity of methane + air was measured. The laminar burning velocity was determined at 298 K and equivalence ratios 0.7–1.3. The “real” burning velocity of acetic acid + air was then recalculated from the measured value by the ratio of the reference burning velocity of CH_4 and of that measured on the burner “E2”: $S_{L, \text{real}} = S_{L, \text{acid}} * S_{L, \text{CH}_4, \text{ref}} / S_{L, \text{CH}_4, \text{E2}}$. This procedure turned out to be even more important for the data processing because it was observed that the laminar burning velocity of both CH_4 and of acetic acid was slightly lower each experimental day as illustrated in Supplemental material, Fig. S1. Nevertheless, the procedure of S_L recalculation using two values of the methane burning velocity brought consistent results when comparing measurements performed on two different setups and for the measurements repeated on the same setup after several days, as will be shown in Results section. After each experimental session the installation was flushed with ethanol and the flames of ethanol were also controlled. They behave similarly to the flames of methane with day-to-day variations and significantly lower values of S_L on the burner “E2”. Methane flames were preferred as the reference for recalculation since they are investigated in many laboratories and their burning velocities are well established.

After finalizing the experiments with acetic acid the burner “E2” was disassembled and a green oxidation layer, patina, was found covering the insides of the plenum chamber and the burner head as illustrated in Supplemental material, Figs. S2 and S3. One may conclude that some portion of acetic acid reacted with the inside surface of the burner, as expected. To evaluate the impact of the surface reactions on the burning velocity experiments, an attempt to measure its products was made. With a typical flow of acetic acid + air mixture through the burner, yet without a flame, the gases were collected using a funnel placed above the burner plate and analyzed by a commercial flue gas analyzer (Varioplus Industrial). No CO_2 was detected (detection limit is 5000 ppm), while single digit ppm levels of CO were measured. Therefore the modification of the fresh mixture composition due to the surface reactions could be neglected. It is also important to note that in the experiments of Leplat and Vandooren [11] the burner was made of brass as well. Yet, no noticeable consumption of acetic acid or formation of unexpected products close to the burner surface was reported.

The measurements of the burning velocity of acetic acid + air flames span over 3 non-consecutive days using two different heat flux setups described earlier [24], yet with the same burner “E2”.

Table 1

Experimental laminar burning velocities of acetic acid + air flames at 338 K, 348 K, and 358 K in cm/s. Values with asterisk were obtained using extrapolation procedure.

ϕ	338 K		348 K			358 K	
	Day 2	Day 3	Day 1	Day 2	Day 3	Day 2	Day 3
0.8		14.85		15.87*		16.84*	17.37*
0.9	18.08	18.46		19.09	19.56*	20.12*	20.95*
0.9					19.55*		
1	19.33	19.78			20.85	21.69*	21.95*
1		20.04	20.80	20.58*		21.79*	
1.1	17.98	19.38	20.28	19.97	20.57*	21.21*	21.83*
1.2	16.34	16.88		17.20*	17.89*	18.81*	19.35*

Initial feasibility experiments were carried out on setup I equipped with a larger evaporator (1200 g/h) since the burning velocity of acetic acid was never measured and relatively high fuel flows were expected. These measurements at 348 K are presented in Table 1 as Day 1 experiments. It turned out that capacity of the smaller evaporator (200 g/h) of setup II is sufficient to cover all flows, and the measurements at three temperatures, 338, 348 and 358 K, were continued during Day 2 and Day 3, see Table 1. Employment of the appropriate size of the evaporator (not too large) improves flow stability and thus the accuracy of the measurements, as discussed elsewhere [19].

All experiments were performed at atmospheric pressure, which varied from 1005 to 1014 hPa, and no correction for its variation was considered. The partial vapor pressure of acetic acid prevented measurements to be performed at initial temperatures below 338 K. To maintain the fuel in the gas phase a heating tube was connected between the evaporator and the plenum chamber. The temperature was set to the same temperature as the plenum chamber. The range of equivalence ratios was from 0.8 to 1.2; leaner and richer flames were highly unstable due to much lower burning velocities. Recent studies using the heat flux burners employed two methods for data evaluation: interpolation and extrapolation [19]; extrapolation is required when non-stretched flames become unstable for the flow rates exceeding the burning velocity at smaller differences between the temperature of the plenum chamber and of the burner head, 368 K in the present paper. Extrapolation procedure causes somewhat higher experimental uncertainties, as was thoroughly discussed elsewhere [19,22]. These uncertainties were taken into account in the present data processing. As is indicated in Table 1, all results at 338 K were obtained by interpolation. Most of the combinations of the temperature and equivalence ratio were re-visited twice to understand and interpret unusual day-to-day variations of the burning velocity; further analysis of the data consistency is given in Section 5.

3. Reaction mechanism

The present model is based on the Konnov mechanism version 0.6 [27]. It is, however, significantly different due to recent changes in the backbone hydrogen and C1 chemistry. Hydrogen [28] and syngas [29] reactions have been reviewed and these sub-mechanisms were updated and thoroughly validated. In addition 82 reversible C1 reactions were reviewed and updated focusing on combustion chemistry of formaldehyde and methanol [17]. The original model [27], however, did not include reactions of acetic acid or detailed sub-mechanism for ketene, which were considered in the later editions of the model [8,30,31].

It is well known that ketene, CH_2CO , is a key intermediate in acetic acid combustion, generated during decomposition of the fuel [32] or in flames [11]. It has also been shown to play a major part in the consumption of acetylene, C_2H_2 , and to balance the for-

mation of C_1 and C_2 species in C_2H_2 flames [33,34]. Ketene and ketylenyl radical, HCCO , are important intermediates in combustion of other oxygenates, e.g. acetone [30,31]. Therefore, the focus of the present analysis was on reactions of ketene and of acetic acid as described in the following. The rate constant expressions in the present model are based on experimental, theoretical or review data and there has been no modification (adjustment or tuning) of the values to accommodate experimental results. Rate constants have units of $\text{cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ and the activation energies are in cal/mol . The mechanism consists of 1140 reactions and 100 species. Due to the size of the mechanism it is not presented as a whole here, however, selected important reactions and their rate constants are discussed in the text and summarized in Table 2.

3.1. Reactions of CH_2CO

Wagner and Zabel [35] investigated thermal decomposition of ketene



in a wide range of pressures and derived low- and high-pressure limiting rate constants, which are accepted in the present paper together with collisional efficiencies suggested by Warnatz [36]. Effective second-order rate constant also provided by Wagner and Zabel [35] was later substantiated in shock tube experiments by Mackie and Doolan [37], by Frank et al. [38], and by Friedrichs and Wagner [39]. Hidaka et al. [40] found it necessary to increase the rate suggested by Wagner and Zabel [35] by 10%, which is well within the experimental uncertainties and typical data scattering of a factor of 2. Frank et al. [38] considered another channel of ketene decomposition forming $\text{HCCO} + \text{H}$ and evaluated its rate as $4\text{E} + 8 \text{ cm}^3/\text{mol s}$ within 2000–2300 K temperature range. This channel is not included in the present mechanism since C–H bond is much stronger than C–C bond.

Major products of the reaction between ketene and hydrogen atoms



are methyl radical and CO. Hranisavljevic et al. [41] summarized earlier experimental studies of this reaction by Carr et al. [42], Michael et al. [43], Umemoto et al. [44], Frank et al. [38], performed experiments over 863–1400 K range and TST calculations and derived the rate constant which was then recommended by Baulch et al. [45]. Subsequent theoretical studies of Joshi et al. [46], Lee and Bozzelli [47], and Senosiain et al. [48] are all consistently agree with experiments as well. The most recent rate constant [48] is adopted in the present paper with the estimated uncertainty of a factor of 1.5.

The rate constant of the channel



was derived by Hidaka et al. [40] to fit H_2 concentration profiles during ketene pyrolysis; they suggested activation energy of 8.6 kcal/mol. Lee and Bozzelli [47] performed ab initio calculations of this system and found a barrier of 12.3 kcal/mol to form ketylenyl radical and H_2 . Thus the rate constant was estimated using this barrier in the present paper with an uncertainty of a factor of 3. Carl et al. [49] measured the rate constant between HCCO and H_2 within 438–761 K. Their rate constant is higher than the reverse rate of reaction (3) calculated from the rate of Hidaka et al. [40] by a factor 2–5. One may note, however, that reaction between ketylenyl radical and H_2 may have other products, e.g. CHCHOH or $\text{CH}_3 + \text{CO}$.

Table 2

Selected reactions of ketene and acetic acid; units are cm³, mol, s, cal and K, $k = AT^n \exp(-E_a/RT)$. UF – uncertainty factor.

No	Reaction	A	n	E _a	T range, K	UF	Source
1	CH ₂ CO(+M)=CH ₂ +CO(+M)	3.00E+14	0	71,000	1300–2000	2	[35]
	Low pressure limit:	3.00E+15	0	57,000	1300–2000	2	[35]
	Enhanced third-body efficiencies (relative to Ar): H ₂ =2.5, H ₂ O=6.2, CO=1.9, CO ₂ =3.8, CH ₄ =3.2, CH ₃ OH=7.5						[36]
2	CH ₂ CO+H=CH ₃ +CO	7.77E+08	1.45	2780	200–2500	1.5	[48]
3	CH ₂ CO+H=HCCO+H ₂	1.00E+14	0	12,300	300–2000	3	See text
4	CH ₂ CO+O=HCO+HCO	1.80E+11	0	1350	300–1000	10	[45]
5	CH ₂ CO+O=HCO+CO+H	1.80E+11	0	1350	300–1000	10	[45]
6	CH ₂ CO+O=CH ₂ O+CO	3.60E+11	0	1350	300–1000	10	[45]
7	CH ₂ CO+O=CH ₂ +CO ₂	1.08E+12	0	1350	300–1000	10	[45]
8	CH ₂ CO+O=HCCO+OH	1.87E+14	0	16,690	500–2000	3	[54]
9	CH ₂ CO+OH=CH ₂ O+HCO	1.00E+11	0	0	300–3000	3	See text
10	CH ₂ CO+OH=CH ₂ OH+CO	2.64E+12	0	–550	300–3000	3	See text
11	CH ₂ CO+OH=HCCO+H ₂ O	1.78E+13	0	5840	300–3000	3	See text
12	CH ₂ CO+OH=CH ₃ +CO ₂	3.00E+11	0	0	300–3000	3	See text
13	CH ₂ CO+OH=CH ₂ CO ₂ H	4.00E+12	0	0	300–3000	10	See text
14	CH ₂ CO+CH=C ₂ H ₂ +CO+H	1.232E+14	0	0	295	3	See text
15	CH ₂ CO+CH=C ₂ H ₃ +CO	2.175E+13	0	0	295	3	See text
16	CH ₂ CO+CH ₂ =HCCO+CH ₃	3.60E+13	0	11,000	1100–1920	3	[40]
17	CH ₂ CO+CH ₂ =C ₂ H ₄ +CO	1.60E+04	2.8	6210	200–3000	3	[67]
18	CH ₂ CO+SCH ₂ =C ₂ H ₄ +CO	6.00E+13	0	–526	300–462	2	[69]
19	CH ₂ CO+CH ₃ =C ₂ H ₅ +CO	2.40E+12	0	8000	300–2000	5	[71]
20	CH ₂ CO+CH ₃ =HCCO+CH ₄	7.50E+12	0	13,000	1100–1920	3	[40]
21	CH ₂ CO+O ₂ =CH ₂ O+CO ₂	2.00E+13	0	61,500	300–2000	10	[73]
22	CH ₂ CO+HO ₂ =CH ₂ O+CO+OH	6.00E+11	0	12,738	300–2000	10	[73]
23	CH ₂ CO+C ₂ H ₃ =C ₃ H ₅ +CO	1.00E+12	0	3000	300–2000	10	[73]
24	HCCO+H=SCH ₂ +CO	1.20E+14	0	0	280–2000	1.5	[45]
25	HCCO+H=CH ₂ +CO	1.05E+13	0	0	280–2000	1.5	[45]
26	HCCO+O=CO+CO+H	1.00E+14	0	0	280–2500	2	[45]
27	HCCO+O=CH+CO ₂	2.95E+13	0	1110	280–1000	2	[45,77]
28	HCCO+OH=HCCOH+O	2.95E+12	0	31,138	500–2000	3	[54]
29	HCCO+OH=C ₂ O+H ₂ O	1.43E+04	2.65	1472	500–2000	3	[54]
30	HCCO+OH=CO+HCOH	2.76E+13	0.09	–20	500–2000	3	[54]
	+	2.86E+12	0.37	–24	500–2000	3	[54]
31	HCCO+OH=CO+CH ₂ O	3.76E+13	–0.06	156	500–2000	3	[54]
32	HCCO+OH=H+CHOCO	2.63E+08	1.41	849	500–2000	3	[54]
33	HCCO+OH=CO ₂ +CH ₂	7.08E+19	–2.3	824	500–2000	3	[54]
34	HCCOH+H=H+CH ₂ CO	1.00E+14	0	4000	300–2000	3	See text
35	HCCO+O ₂ =CO ₂ +CO+H	4.76E+12	–0.142	1150	300–2500	2	[83]
36	HCCO+O ₂ =CO+CO+OH	1.90E+11	–0.02	1023	300–2500	2	[83]
37	HCCO+O ₂ =O+CHOCO	2.17E+02	2.692	3541	300–2500	2	[83]
38	HCCO+CH=CO+C ₂ H ₂	5.00E+13	0	0	300–2000	5	[80]
39	HCCO+CH ₂ =C ₂ H+CH ₂ O	1.00E+13	0	2000	300–2000	5	[53]
40	HCCO+CH ₂ =C ₂ H ₃ +CO	3.00E+13	0	0	300–2000	5	[53]
41	HCCO+CH ₃ =C ₂ H ₄ +CO	2.00E+12	0	0	300–2000	5	[40]
42	HCCO+C ₂ H ₂ =C ₃ H ₃ +CO	1.00E+11	0	3000	300–2000	5	[72]
43	HCCO+HCCO=CO+C ₂ H ₂ +CO	1.00E+13	0	0	300–2000	5	[80]
44	CH ₃ CO ₂ H=CH ₄ +CO ₂	3.10E+13	0	70,300	500–2000	3	[100]
45	CH ₃ CO ₂ H=CH ₂ CO+H ₂ O	4.93E+13	0	75,700	500–2000	3	[100]
46	CH ₃ CO ₂ H=CH ₂ C(OH) ₂	5.54E+12	0	68,900	500–2000	3	[100]
47	CH ₂ C(OH) ₂ =CH ₂ CO+H ₂ O	1.04E+13	0	45100	500–2000	3	[100]
48	CH ₂ C(OH) ₂ +H=CH ₃ +HCOOH	3.00E+12	0	0	300–2000	10	See text
49	CH ₂ C(OH) ₂ +O=CH ₂ OH+HOCO	1.00E+13	0	0	300–2000	10	See text
50	CH ₂ C(OH) ₂ +OH=CH ₃ OH+HOCO	1.00E+12	0	0	300–2000	10	See text
51	CH ₃ CO+OH=CH ₂ CO+H ₂ O	2.80E+13	0	0	300–2000	3	[104]
52	CH ₃ CO+OH=CH ₄ +CO ₂	1.40E+13	0	0	300–2000	3	[104]
53	HOCO+CH ₃ =CH ₂ CO+H ₂ O	2.80E+13	0	0	300–2000	10	See text
54	HOCO+CH ₃ =CH ₄ +CO ₂	1.40E+13	0	0	300–2000	10	See text
55	CH ₃ CO+OH+M=CH ₃ CO ₂ H+M	1.00E+22	–2.0	0	300–2000	10	See text
56	HOCO+CH ₃ +M=CH ₃ CO ₂ H+M	1.00E+22	–2.0	0	300–2000	10	See text
57	CH ₃ CO ₂ H+H=CH ₂ CO ₂ H+H ₂	1.68E+04	2.9	7376	500–2000	5	[16]
58	CH ₃ CO ₂ H+H=CH ₃ CO ₂ +H ₂	1.79E+06	2.3	15,215	500–2000	5	[16]
59	CH ₃ CO ₂ H+O=CH ₂ CO ₂ H+OH	2.30E+04	2.93	7040	250–2000	10	See text
60	CH ₃ CO ₂ H+O=CH ₃ CO ₂ +OH	8.50E+05	2.1	9880	300–2000	10	See text
61	CH ₃ CO ₂ H+OH=CH ₃ CO ₂ +H ₂ O	2.12E–04	4.8	–3949	500–2000	2	See text
62	CH ₃ CO ₂ H+OH=CH ₂ CO ₂ H+H ₂ O	4.40E+04	3.8	–725	500–2000	2	See text
63	CH ₃ CO ₂ H+HO ₂ =CH ₂ CO ₂ H+H ₂ O ₂	4.02E–06	5.3	15,153	500–2000	5	[16]
64	CH ₃ CO ₂ H+HO ₂ =CH ₃ CO ₂ +H ₂ O ₂	1.41E–02	4.2	20,862	500–2000	5	[16]
65	CH ₃ CO ₂ H+CH ₃ =CH ₂ CO ₂ H+CH ₄	2.14E–01	4.0	8222	500–2000	5	[16]
66	CH ₃ CO ₂ H+CH ₃ =CH ₃ CO ₂ +CH ₄	9.30E+01	3.0	10,773	500–2000	5	[16]
67	CH ₃ CO ₂ H+O ₂ =CH ₂ CO ₂ H+HO ₂	1.26E+13	0	49,043	500–2000	10	[13]
68	CH ₃ CO ₂ =CH ₃ +CO ₂	2.196E+11	0.294	4580	200–3000	3	[117]
69	CH ₂ CO ₂ H+H=CH ₂ CO+H ₂ O	1.00E+14	0	0	300–2000	10	See text
70	CH ₃ CO ₂ H+H=CH ₂ CO+H ₂ O	1.00E+14	0	0	300–2000	10	See text

It is generally accepted that reaction between ketene and O atoms proceeds via addition followed by fragmentation of the excited adduct [42,45] and has several product channels:



Recommendations of Baulch et al. [45] of the total rate and branching ratios based on the study of Washida et al. [50] are adopted here. Later theoretical calculations of Sun et al. [51] supported reaction (7) as the major product channel. Hidaka et al. [52] suggested much higher rate constant of reaction (7) to model their shock tube experiments of ketene oxidation. Uncertainties of these rate constants are about a factor of 2 at room temperature rising to a factor of 10 at 1000 K [45]. Hidaka et al. [52] attempted inclusion of direct abstraction reaction



with the rate constant estimated by Miller et al. [53], yet did not find it important. In the present paper the rate constant of reaction (8) calculated by Xiong et al. [54] is adopted with the uncertainty of a factor of 3.

Baulch et al. [45] discussed four channels of the reaction between ketene and OH radical



Total recommended rate constant [45] was based on the negative activation energy observed experimentally from 193 to 423 K by Brown et al. [55], and extended up to 1000 K. Branching ratios determined at room temperature [56] were accepted by Baulch et al. [45]. Higher rate constant for reaction (9) derived by Vandooren and Van Tiggelen [57] was not considered due to indirect and simplified approach. Hou et al. [58] performed ab initio calculations and found calculated branching ratios in agreement with Grussdorf et al. [56]. Overall temperature-independent rate constant of Baulch et al. [59] is adopted in the present paper together with the calculated branching ratios [58]. Negative temperature dependence observed experimentally [55] and endorsed by Baulch et al. [45] is then mostly due to channel (10). Reaction channel stabilizing $\text{CH}_2\text{CO}_2\text{H}$



is also included based on the calculations of Hou et al. [58] for atmospheric pressure and low temperatures since its reverse describes decomposition important for acetic acid oxidation. Temperature-independent rate is assumed even though Hou et al. [58] found that the branching ratio rapidly decreases at higher temperatures due to lack of stabilization of excited $\text{CH}_2\text{CO}_2\text{H}$ formed in recombination. To check the validity of this rate, its reverse was calculated. The activation energy of decomposition reaction (–13) was found to be about 55 kcal/mol, close to the estimated value by Gasnot et al. [13] (58 kcal/mol) and is much different from the value suggested by Leplat and Vandooren [11] (29

kcal/mol). Note that the barrier of $\text{CH}_2\text{CO}_2\text{H}$ decomposition into ketene and OH calculated by Hou et al. [58] is 46 kcal/mol. In the view of many uncertainties this rate constant is probably accurate to within a factor of 10. Thermodynamic properties of $\text{CH}_2\text{CO}_2\text{H}$ radicals were calculated using Thergas software [60]. The enthalpies of its formation were found close to literature data [61,62].

Reaction between ketene and ground state CH radical has two major channels



and minor channels forming $\text{H}_2\text{C}_2 + \text{HCO}$ and $\text{C}_2\text{H}_2 + \text{HCO}$ as revealed by theoretical study [63]. Canosa-Mas et al. [64] estimated the total rate to be about 1.26×10^{14} and suggested branching ratio favoring reaction (15). Frey and Walsh [65] re-visited modeling of their experimental data [64] and suggested 85% branching ratio for reaction (14). This was essentially supported by Li et al. [63]. The total rate measured by Hancock and Heal [66] is very close to the earlier estimation [64] and is adopted here. Large uncertainty of a factor of 3 is attributed to the branching ratio, while the total rate is probably accurate to within a factor of 1.5.

Hidaka et al. [40] suggested the rate constant of reaction



to match CH_4 yield in their experiments on ketene pyrolysis within 1102–1921 K range, while for reaction



they accepted the rate constant assumed by Frank et al. [38] at 2200 K. In fact, Frank et al. [38] concluded that reaction (17) has negligible role under their experimental conditions, and does not affect H and CO profiles recorded in a shock tube. Later, Lu et al. [67] performed ab initio calculations of this rate constant, which is accepted in the present paper. In the absence of direct measurements, the rate constants are probably accurate within a factor of 3.

Collisional and reactive removal of singlet methylene, SCH_2 , in



has been studied in many works, yet it was always difficult to separate both processes. Hancock and Heal [68] found that the rate of removal decreases with temperature increase up to 859 K. Hayes et al. [69] confirmed this observation and derived an Arrhenius rate constant which is adopted in the present paper. The rate of collisional deactivation is assumed to be temperature-independent. In these assumptions at 298 K the yield of CO in reaction of ketene with singlet methylene is about 80%, which could be compared with the value $63 \pm 13\%$ experimentally obtained by Lovejoy et al. [70]. Remaining differences between these studies [68–70] indicate uncertainty of a factor of 2.

The rate constant of reaction



was first estimated by Borisov et al. [71] as $2.4 \times 10^{12} \exp(-8 \text{ kcal/mol/RT})$ in their study of high temperature acetaldehyde oxidation. Woods and Haynes [33] tested the mechanism of Miller and Melius [72] analyzing concentration profiles of hydrocarbons in rich flames and suggested that reaction (19) should be incorporated into the model to balance formation of C1 and C2 species.

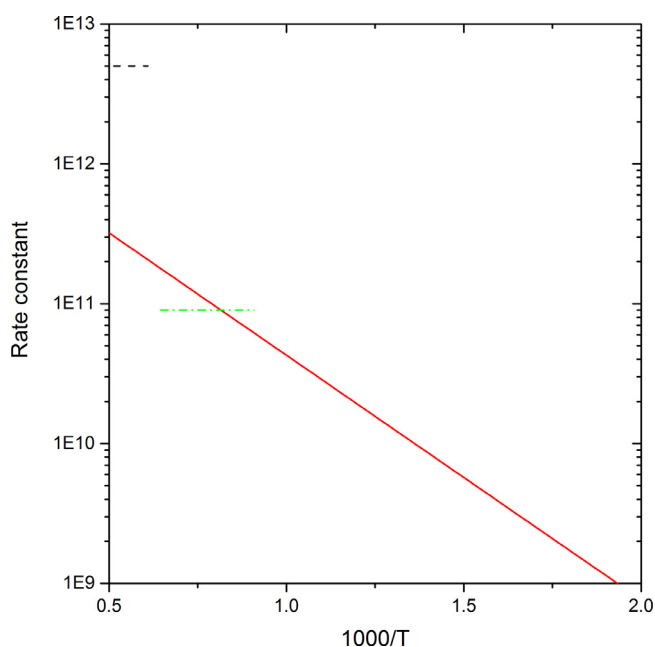


Fig. 1. Rate constants of reaction $\text{CH}_2\text{CO} + \text{CH}_3 = \text{C}_2\text{H}_5 + \text{CO}$. Red solid line: Borisov et al. [71]; black dashed line: Woods and Haynes [33]; green dash-dot line: Hidaka et al. [40]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The estimated rate constant [33] was $5\text{E}+12 \text{ cm}^3/\text{mol s}$ ($\pm 50\%$) within 1640–1950 K, which is much higher than the value suggested later by Hidaka et al. [40] ($9\text{E}+10 \text{ cm}^3/\text{mol s}$ within 1100–1920 K) to fit profiles of C_2H_4 during ketene pyrolysis. Woods and Haynes [33] admitted that their rate constant is probably too high due to expected activation barrier of CH_3 addition to alkenes around 8 kcal/mol. One may note that other competing reactions, e.g. (17) were not considered in these models [33,72]. Leplat and Vandooren [11] adopted the rate constant of Woods and Haynes [33] arguing for its important role in the formation of C2 species in flames. Further analysis shows that the rate constant suggested by Hidaka et al. [40] was derived by comparing modeling with the experiments using the mixture of 2.2 % CH_2CO + Ar which was investigated from 1100 to 1550 K. In this temperature range their value is close to the expression proposed by Borisov et al. [71] as shown in Fig. 1. The uncertainty of this expression adopted in the present paper is probably of a factor of 5.

The rate constant of competing channel



was derived by Hidaka et al. [40] to fit CH_4 concentration profiles during ketene pyrolysis. It is accepted here with uncertainty of a factor of 3.

The rate constants of the following reactions



have never been measured. Provisionally the rates suggested by Tan et al. [73] are adopted in the present paper with uncertainty of a factor of 10.

3.2. Reactions of HCCO

Determinations of the overall rate between ketylenyl radicals and H atoms are consistent both at room [74,75] and at high [76] temperatures. Recommendation of Baulch et al. [45] is adopted together with the branching ratio indicating dominant formation of singlet methylene [75] and uncertainty of about a factor of 1.5



Similar consistency was observed for the major channel of reaction between ketylenyl radicals and O atoms



at low [74] and high [76] temperatures. In the second channel



both ground state and quartet CH ($a^4\Sigma^-$) could be formed [77]. They were not distinguished experimentally and not separated in the present model. Recommendation of Baulch et al. [45] are based on these studies and adopted here with uncertainty of a factor of 2.

Reaction between ketylenyl and hydroxyl radicals has many product channels both on triplet and on singlet surfaces as was recently demonstrated in theoretical analyses [54,78]. Both studies found that primary products are CO and singlet or triplet HCOH. In the present mechanism the singlet HCOH is not included, thus it is assumed that it is rapidly quenched to the triplet state. Formation of adducts, such as singlet and triplet CHCOOH , hydroxyketene HOCHCO , and their isomerization forming glyoxal $(\text{HCO})_2$ was also analyzed, yet these reactions are hardly important at moderate pressures and are not included in the present model. For consistency the rate constants calculated by Xiong et al. [54] were adopted here with uncertainty of a factor of 3 for the following channels:



Hydroxyacetylene HCCOH is formed in one of several possible channels in reaction between acetylene and OH radicals [79]. The chemistry of HCCOH, however, is basically unknown. Miller and Bowman [80] suggested that hydroxyacetylene is converted into ketene in reaction



which essentially mimics another product channel of $\text{C}_2\text{H}_2 + \text{OH}$. Therefore, these two reactions of formation and consumption of HCCOH can be excluded from kinetic model, if the rate of reaction $\text{C}_2\text{H}_2 + \text{OH} = \text{CH}_2\text{CO} + \text{H}$ is adjusted appropriately. However, another possible reaction (–28) mentioned above competes for HCCOH radicals. The rate of reaction (34) was estimated by Miller and

Bowman [80] as $1\text{E}+13\text{ cm}^3/\text{mol s}$ and then used in many kinetic schemes. Senosiain et al. [81] and Carl et al. [82] found that interaction between HCCOH and H atoms has entrance energy barriers. It was also suggested [82] that addition of H atoms to terminal carbon is the preferential pathway with the barrier of 4 kcal/mol. The rate constant estimated in the present paper using this barrier has uncertainty not better than a factor of 3.

Baulch et al. [45] did not consider the theoretical analysis of Klippenstein et al. [83] and recommended only overall rate constant of reaction between ketenyl radical and molecular oxygen. Calculated rate constants [83] are adopted in the present paper since they are in agreement with the total rate measured by Carl et al. [84] and were further substantiated experimentally [85,86] in terms of branching ratios. Uncertainty is probably of a factor of 2.



The rate constants of the following reactions have not been measured. Estimations [40,53,72,80] accepted here are often based on assumption of barrierless insertion into C–C bond with predominant decomposition of excited intermediate complex forming CO. These rates are rather uncertain probably within a factor of 5.



3.3. Decomposition and formation of acetic acid

Since early experimental studies in flow [32,87] and static [88] reactors the thermal decomposition of acetic acid has been interpreted in terms of two reactions



The same initiation reactions were observed at high temperatures in shock tube experiments supported by detailed kinetic modeling [37]. Further investigations [89,90] confirmed formation of CH_4 , CO_2 and H_2O as primary products of the dissociation. Reverse reaction between ketene and water (–45) was also substantiated experimentally [91]. Several theoretical studies aimed at understanding and quantification of the decomposition of acetic acid and on hydration reaction of ketene (–45) since it was considered as a major route forming acetic acid in the atmosphere. It was found that reaction of ketene with water and acid decomposition can be catalyzed by additional water [92–94] or acid [93,95] molecules that significantly decrease apparent activation energies of the processes. However, these catalyzed reactions are important at low temperatures only, while at typical combustion conditions the unimolecular decomposition prevails [92]. Moreover, theoretical studies revealed that two mechanisms of acetic

acid dehydration (45) should be considered [96–98]: direct process and two-step mechanism via formation of 1,1-ethenediol



and subsequent dehydration



It was even suggested [99] that in hydration reaction of ketene (–45) formation of ethenediol is energetically slightly preferred to the direct process forming acetic acid. In the present paper the rate constants of reactions (44)–(47) calculated by Clark et al. [100] are adopted with uncertainty of a factor of 3. Recently Elwardany et al. [101] performed shock tube study of acetic acid decomposition and interpreted their results in terms of reactions (44) and (45). They observed pressure dependence from 1 to 6 atm and found their measurements consistently higher (by almost an order of magnitude) than those of Mackie and Doolan [37]. They are also somewhat higher, however much closer to the theoretical expressions of Clark et al. [100] if the rates of reactions (45) and (46) are combined for ketene formation channel. Clearly further theoretical and experimental studies are needed to improve the accuracy of the rate constants for acetic acid decomposition.

Thermodynamic properties of 1,1-ethenediol were calculated using Thergas software [60]. Although these calculations based on Benson's thermochemistry rules cannot be very accurate for small molecules, the enthalpy of ethenediol formation was found close to literature data [102]. The chemistry of 1,1-ethenediol has not yet been studied; in the present paper reactions with O and H atoms and with OH radicals were introduced assuming barrierless insertion into C=C bond and decomposition of adducts:



These estimates and the products themselves are highly uncertain. On the other hand, reactions (48)–(50) were not found sensitive or important at the conditions covered in the present paper.

Reverse reactions (–44) and (–45) are not sufficient to explain acetic acid formation observed in flames by, e.g. Zervas [7], since they have rather high activation energies. Other reactions have been proposed, such as bimolecular recombination of CH_3CO and OH [7,8] or of CH_3 and HOCO [7,14]. Intermediate adducts, however, are highly excited and rapidly decompose into other products. Tsang and Hampson [103] suggested formation of ketene and water



or of $\text{CH}_3 + \text{CO} + \text{OH}$ in reaction of acetyl and hydroxyl radicals. Butkovskaya et al. [89,104] concluded that reaction (51) is favorable over the second major channel of the decomposition of excited acetic acid



with the branching ratio of 2:1. The total rate estimated by Tsang and Hampson [103] is adopted here with an uncertainty of a factor of 3. Similar consideration leads to the estimated rates of reactions



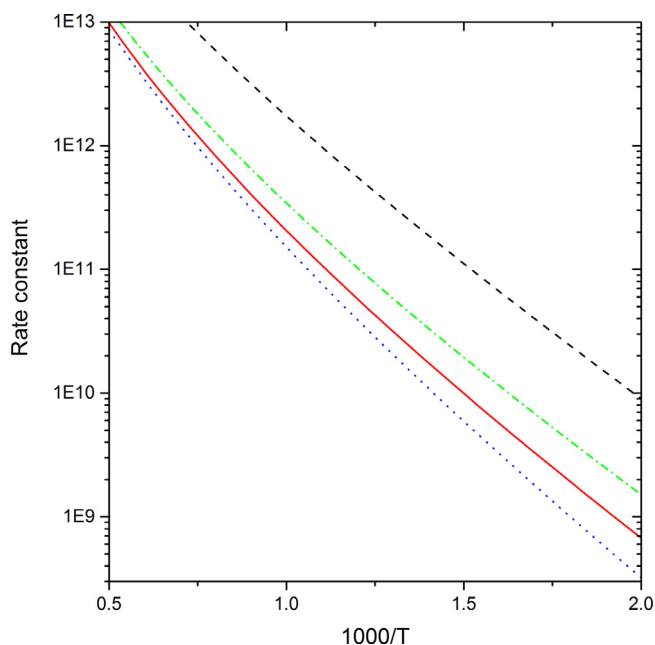


Fig. 2. Rate constants relevant to reaction $\text{CH}_3\text{CO}_2\text{H} + \text{H} = \text{CH}_2\text{CO}_2\text{H} + \text{H}_2$. Red solid line: theoretical by Mendes et al. [16]; black dashed line: estimation by Gasnot et al. [13]; green dash-dot and blue dotted lines: theoretical for methyl acetate + H by Tan et al. [105]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



with higher uncertainty of a factor of 10. Only termolecular reactions



may form acetic acid. The rates of these reactions are estimated also with high uncertainty.

3.4. Reactions of acetic acid

All reactions of $\text{CH}_3\text{CO}_2\text{H} + \text{R}$ have two channels forming $\text{CH}_2\text{CO}_2\text{H}$ or CH_3CO_2 radicals. In the following existing estimations, theoretical calculations and experimental data, if available, are compared to substantiate the choice of particular rate expression. The rate constants relevant to reaction



are shown in Fig. 2. The theoretical expression of Mendes et al. [16] is almost 4 times lower at 2000 K and 10 times lower at 500 K than the estimation of Gasnot et al. [13] employed by Leplat and Vandooren [11]. Tan et al. [105] calculated rate constants of methyl acetate + radicals using two theoretical approaches. These rate constants for reaction $\text{CH}_3\text{CO}_2\text{CH}_3 + \text{H} = \text{CH}_2\text{C}(\text{=O})\text{OCH}_3 + \text{H}_2$ are also depicted in Fig. 2 since one may expect close rates of H atom abstraction from methyl group for acetic acid and methyl acetate. Remarkably, the theoretical expression for acetic acid [16] is found between these two rate constants. On the other hand, the difference between the two levels of theory for methyl acetate is about a factor of 5 at lower temperatures that indicates possible uncertainty of the ab initio calculations.

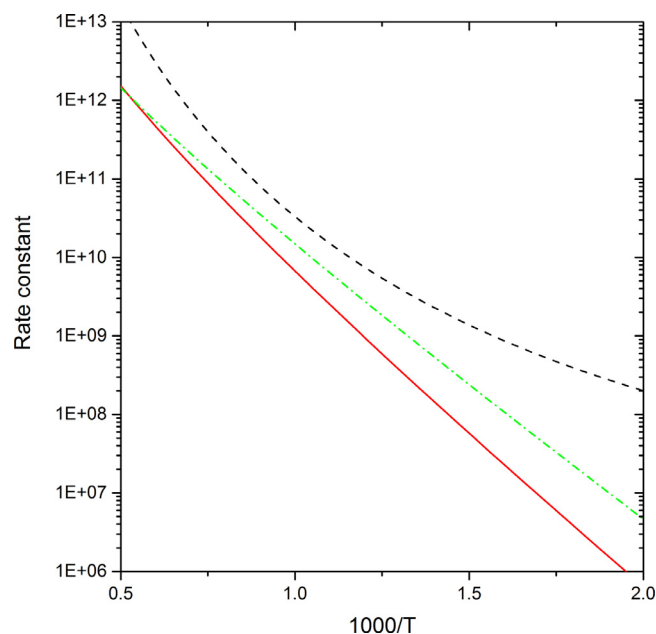


Fig. 3. Rate constants relevant to reaction $\text{CH}_3\text{CO}_2\text{H} + \text{H} = \text{CH}_3\text{CO}_2 + \text{H}_2$. Red solid line: theoretical by Mendes et al. [16]; black dashed line: estimation by Leplat and Vandooren [11]; green dash-dot line: generic rate by Sumathi et al. [106]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The rate constants of reaction



are shown in Fig. 3. The theoretical expression of Mendes et al. [16] is almost an order of magnitude lower than the estimation of Leplat and Vandooren [11] at high temperatures and further deviates at temperatures approaching 500 K. Generic rate constant derived using group additivity rules for H atom abstraction from acids by Sumathi et al. [106] is very close to the ab initio calculations at high temperatures, while deviating at lower temperatures thus indicating remaining uncertainty of the rate constant.

Figure 4 shows the estimation of Gasnot et al. [13] for the rate constant of reaction



and two rate constants calculated by Tan et al. [105] using different levels of theory for methyl acetate + O atoms forming $\text{CH}_2\text{C}(\text{=O})\text{OCH}_3 + \text{OH}$. The spread of these rates is rather large spanning more than two orders of magnitude at lower temperatures. Provisionally the rate constant lying in between two others is selected with the estimated uncertainty of a factor of 10.

The estimation of Leplat and Vandooren [11] for the rate constant of reaction



is shown in Fig. 5. Also presented is the rate constant calculated by Marshall and Glarborg [107] for reaction of formic acid with O atoms: $\text{HCOOH} + \text{O} = \text{HCOO} + \text{OH}$. The similarity in reactivity (if any) between acidic groups of acetic and formic acids was tested by comparing rate constants of reactions $\text{CH}_3\text{CO}_2\text{H} + \text{H} = \text{CH}_3\text{CO}_2 + \text{H}_2$ (58) and of $\text{HCOOH} + \text{H} = \text{HCOO} + \text{H}_2$ also calculated by Marshall and Glarborg [107]. The last rate constant (not shown in Fig. 3) was found 4–5 times lower than that calculated for acetic acid [16] over entire temperature range. Although this trend could be just fortuitous, in the present paper the rate constant for formic acid

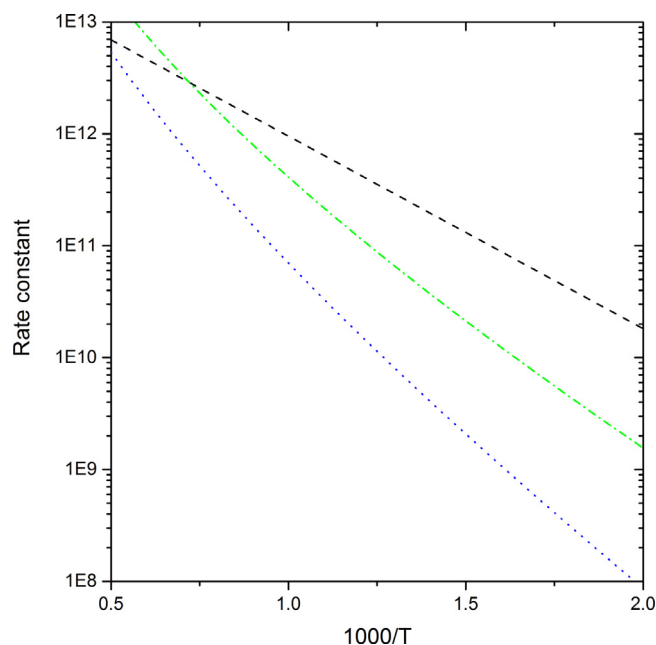


Fig. 4. Rate constants relevant to reaction $\text{CH}_3\text{CO}_2\text{H} + \text{O} = \text{CH}_2\text{CO}_2\text{H} + \text{OH}$. Black dashed line: estimation by Gasnot et al. [13]; green dash-dot and blue dotted lines: theoretical for methyl acetate + O by Tan et al. [105]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

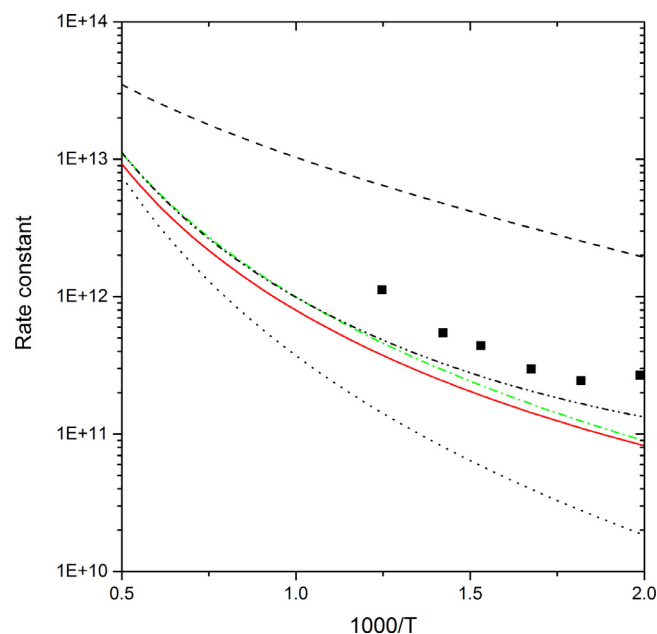
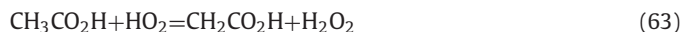


Fig. 6. Rate constants relevant to reaction $\text{CH}_3\text{CO}_2\text{H} + \text{OH} = \text{CH}_2\text{CO}_2\text{H} + \text{H}_2\text{O}$. Red solid line: theoretical by Mendes et al. [16]; black dashed line: estimation by Gasnot et al. [13]; green dash-dot and blue dotted lines: theoretical for methyl acetate + OH by Tan et al. [105]; black dash-double dot line: total rate constant [16]. Symbols: experiment of Khamaganov et al. [110]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



with the first prevailing at around room temperatures. Majority of experiments indicate negative activation energy for the overall rate at low temperature, therefore this rate recommended by DeMore et al. [115] was adopted in the model of Battin-Leclerc et al. [8] and attributed to reaction (61). However, Khamaganov et al. [110] found that above ~ 530 K the overall rate constant increases. Theoretical studies [109,112] showed that abstraction of acidic H (61) indeed dominates at lower temperatures, while at higher temperatures the methyl channel (62) gains its importance. Figure 6 summarizes available experimental data above 500 K [110], theoretical calculations [16] and the rate constant estimated by Gasnot et al. [13] for reaction (62). Also shown are two rate constants calculated for methyl acetate + OH by Tan et al. [105]. The measurements of Khamaganov et al. [110] were performed for the total rate constant; therefore the total rate constant of reactions (61) and (62) [16] is also presented here. The difference between these theoretical calculations and experiments is approximately a factor of 2, therefore the theoretical expressions [16] multiplied by 2 were adopted in the present paper. Figure 7 clearly shows that the low-temperature recommendation for the rate constant of reaction (61) adopted by Leplat and Vandooren [11] cannot be extrapolated to high temperatures. Theoretical studies [16,109,112] all indicate acceleration of both reactions (61) and (62) at combustion temperatures.

Similar to reaction of acetic acid with H atoms (Fig. 2), Fig. 8 shows rate constants relevant to reaction



namely, the estimation of Gasnot et al. [13], the theoretical results of Mendes et al. [16], as well as two rate constants calculated for methyl acetate + HO_2 [105]. The estimated rate constant is consistently higher than ab initio calculations. The difference between theoretical studies is still significant indicating possible uncertainty

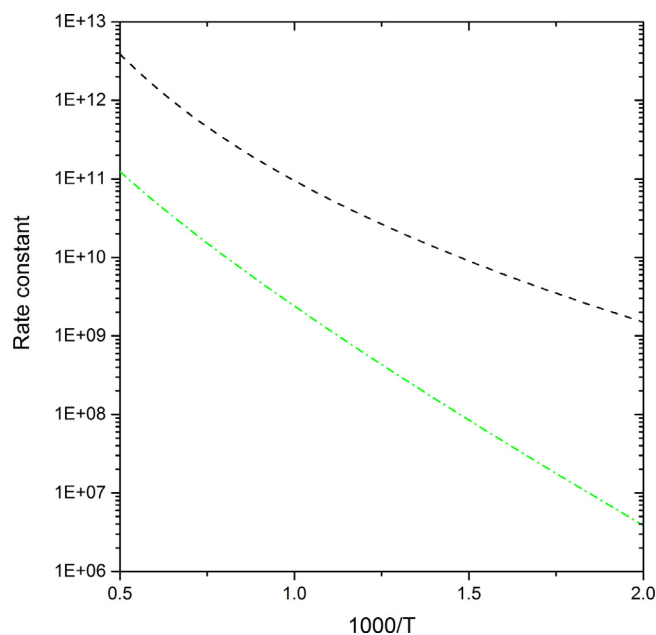


Fig. 5. Rate constants relevant to reaction $\text{CH}_3\text{CO}_2\text{H} + \text{O} = \text{CH}_3\text{CO}_2 + \text{OH}$. Black dashed line: estimation by Leplat and Vandooren [11]; green dash-dot line: theoretical for formic acid + O by Marshall and Glarborg [107]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

multiplied by a factor of 5 was assigned to the reaction of acetic acid with the uncertainty of a factor of 10.

Due to importance in atmospheric chemistry, reaction of acetic acid with OH radicals has been extensively studied, yet mostly at low temperatures [108–111,61,112–114]. It was interpreted in terms of two channels



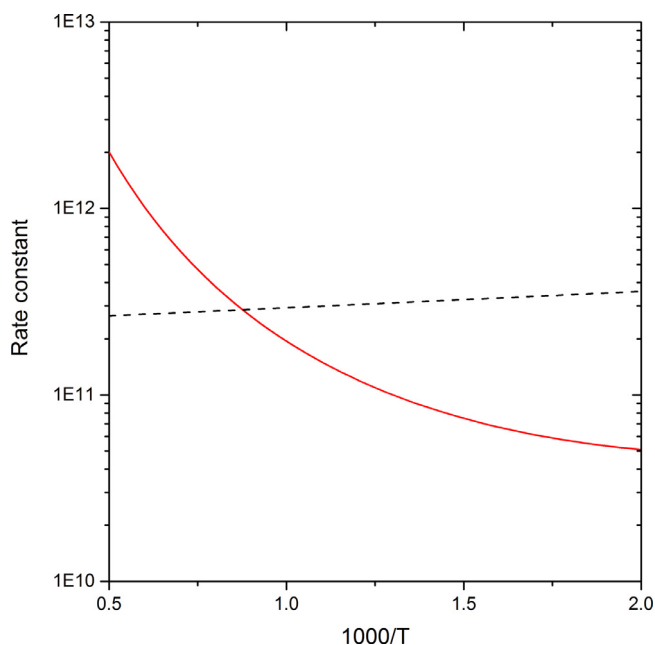


Fig. 7. Rate constants of reaction $\text{CH}_3\text{CO}_2\text{H} + \text{OH} = \text{CH}_3\text{CO}_2 + \text{H}_2\text{O}$. Red solid line: theoretical by Mendes et al. [16]; black dashed line: low-temperature recommendation by DeMore et al. [115]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

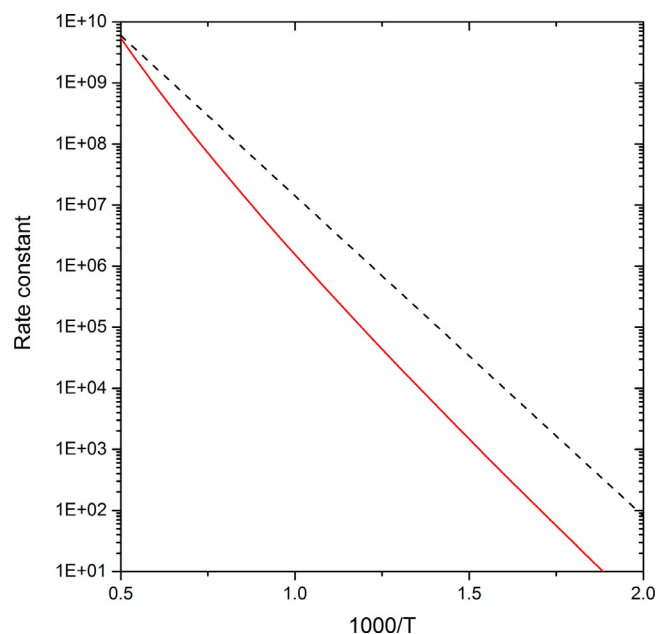


Fig. 9. Rate constants of reaction $\text{CH}_3\text{CO}_2\text{H} + \text{HO}_2 = \text{CH}_3\text{CO}_2 + \text{H}_2\text{O}_2$. Red solid line: theoretical by Mendes et al. [16]; black dashed line: estimation by Leplat and Vandoooren [11]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

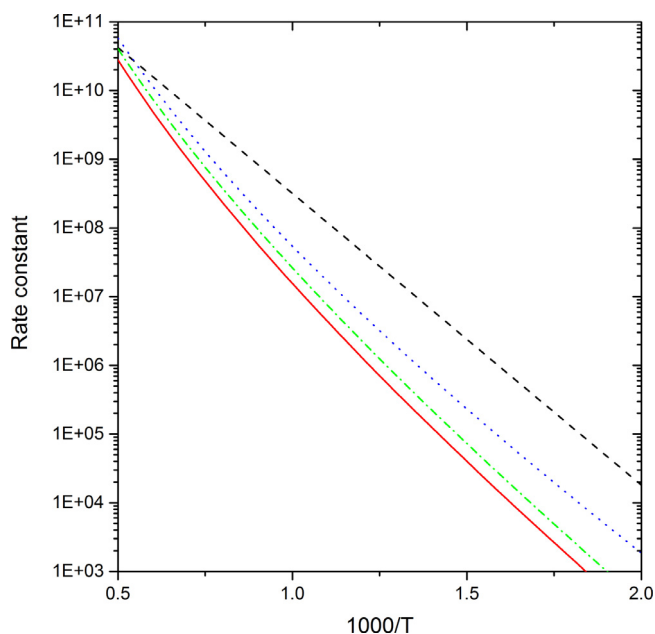


Fig. 8. Rate constants relevant to reaction $\text{CH}_3\text{CO}_2\text{H} + \text{HO}_2 = \text{CH}_2\text{CO}_2\text{H} + \text{H}_2\text{O}_2$. Red solid line: theoretical by Mendes et al. [16]; black dashed line: estimation by Gasnot et al. [13]; green dash-dot and blue dotted lines: theoretical for methyl acetate + HO_2 by Tan et al. [105]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

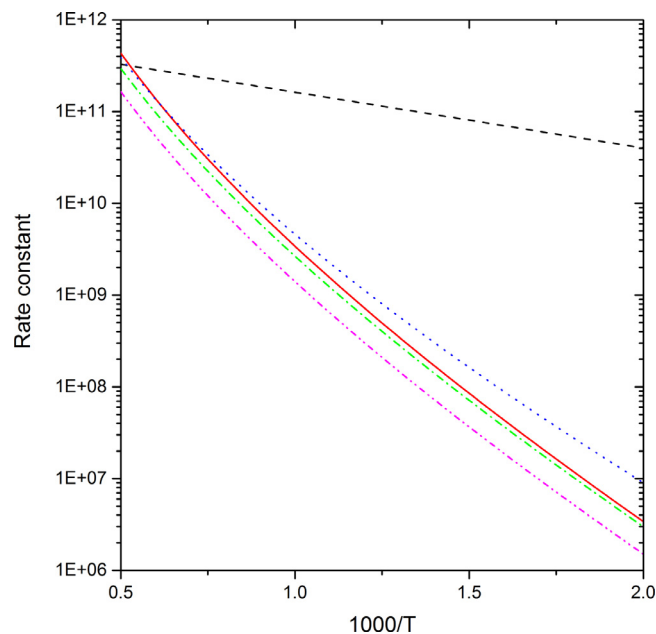


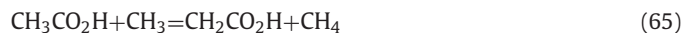
Fig. 10. Rate constants relevant to reaction $\text{CH}_3\text{CO}_2\text{H} + \text{CH}_3 = \text{CH}_2\text{CO}_2\text{H} + \text{CH}_4$. Red solid line: theoretical by Mendes et al. [16]; black dashed line: estimation by Gasnot et al. [13]; green dash-dot and blue dotted lines: theoretical for methyl acetate + CH_3 by Tan et al. [105]; pink dash-double dot line: theoretical for methyl acetate + CH_3 by Peukert et al. [116]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

of about a factor of 5. Figure 9 is another illustration of large difference between the estimation of Leplat and Vandoooren [11] for reaction



and the rate constant derived by ab initio calculations [16].

The rate constants relevant to reaction



are summarized in Fig. 10. In addition to the theoretical expression of Mendes et al. [16] for acetic acid, also shown are two rate constants calculated with different levels of theory for methyl acetate + CH_3 [105] and the same rate constant independently calculated by Peukert et al. [116]. All theoretical results for H atom

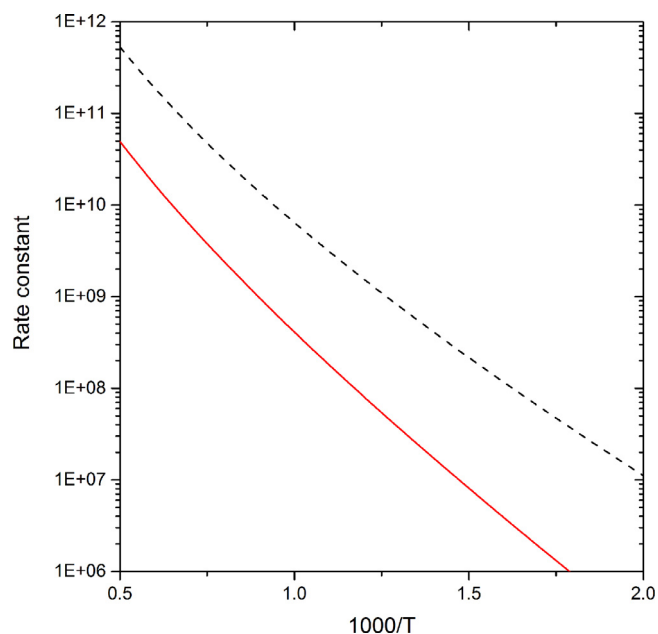


Fig. 11. Rate constants for reaction $\text{CH}_3\text{CO}_2\text{H} + \text{CH}_3 = \text{CH}_3\text{CO}_2 + \text{CH}_4$. Red solid line: theoretical by Mendes et al. [16]; black dashed line: estimation by Leplat and Vandooren [11]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

abstraction from methyl group for acetic acid and methyl acetate presented in Figs. 2, 8, and 10 are consistent showing that close rates could indeed be expected, yet with remaining uncertainty of about a factor of 5. The rate constant implemented by Leplat and Vandooren [11] is apparently a mistake taken from Gasnot et al. [13], who, in turn, converted the activation energy suggested by Warnatz [15] from joules to calories, yet presented it in joules.

Figure 11 presents the estimated [11] and theoretical [16] rate constants for reaction



Comparison of the estimations by Leplat and Vandooren [11] and ab initio calculations shown in Figs. 3, 7, 9, and 11 strongly indicates that assumption of the similar reactivity of different radicals in H atom abstraction from acidic group forming CH_3CO_2 and those from alcohol group [11] could be misleading and typically overestimates the rate constants by more than an order of magnitude.

Reaction of initiation between acetic acid and oxygen was suggested by Salooja [10]



arguing that C–H bond in acid is weaker than O–H bond. Gasnot et al. [13] adopted generic rate constant suggested by Warnatz [15] for reactions with primary H atoms of hydrocarbons. In the absence of other estimates, this rate constant is also adopted in the present paper with uncertainty of a factor of 10.

At atmospheric conditions CH_3CO_2 radicals do not participate in other reactions and, as generally accepted, decompose



The rate constant of this reaction is now known to have rather low barrier of about 5 kcal/mol [58,110,117], which is much smaller than the value used by Battin-Leclerc et al. [8] and by Leplat and Vandooren [11] (14.4 kcal/mol). The rate constant calculated

by Zhou et al. [117] is adopted here with uncertainty of a factor of 3.

$\text{CH}_2\text{CO}_2\text{H}$ radicals are much more stable since decomposition



has an activation energy of about 55 kcal/mol, as discussed above. Thus they may react with oxygen forming $\text{O}_2\text{CH}_2\text{CO}_2\text{H}$ as discussed by Rosado-Reyes and Francisco [61] for atmospheric conditions and by Salooja [10] in relation to oxidation at intermediate temperatures below about 900 K. The radicals $\text{O}_2\text{CH}_2\text{CO}_2\text{H}$ are then decompose forming $\text{CO}_2 + \text{CH}_2\text{O} + \text{OH}$. da Silva [62] calculated the rate constants of formation and decomposition of $\text{O}_2\text{CH}_2\text{CO}_2\text{H}$ and found that at temperatures above 1000 K the rate of recombination with O_2 decreases by several orders of magnitude as compared to room temperature, therefore these reactions were not included in the present model. However, $\text{CH}_2\text{CO}_2\text{H}$ radicals may react with other radicals available in flames. For instance, reaction



may proceed via excited 1,1-ethenediol or acetic acid, which decompose primarily to ketene and H_2O . Similarly, radicals CH_3CO_2 may react with H atoms



The rate constants attributed to reactions (69) and (70) are upper limit estimates with high uncertainty. One should note that reverse reactions (–69) and (–70) do not contribute to ketene hydration since they are energetically much less favorable comparing to reactions (–45) and (–47).

Summarizing the current knowledge of the acetic acid chemistry at high temperatures, one may conclude that many rate constants are known with rather high uncertainty. No attempt to “adjust” the reaction rate expressions was made in the present paper; however, possible modifications within the uncertainty factors listed in Table 2 will be discussed below in connection with the analysis of the mechanism performance.

4. Modeling details

The laminar burning velocities have been calculated with Chemkin IV [118] using the premixed laminar flame speed calculation module. The adaptive grid parameters $\text{GRAD} = 0.03$ and $\text{CURV} = 0.03$ were used, to ensure grid-independent solutions, with typical number of grid points above 500. Species profiles in low-pressure flames [11] have been modeled with the burner-stabilized flame module of Chemkin IV using experimental temperature profile. Multicomponent transport and thermal diffusion options were taken into account in both cases.

5. Results

5.1. Laminar burning velocity

The laminar burning velocities of acetic acid + air flames determined in the present paper are depicted in Figs. 12–14. Uncertainties of the measurements shown as smaller error bars have been evaluated using protocols fully described by Alekseev et al. [19] and include, where appropriate, additional uncertainty due to extrapolation procedure as mentioned above. Note that at initial temperature of 348 K the measurements were performed at two different setups, yet with the same burner. Most of equivalence ratios at each temperature were visited twice to understand and interpret unusual day-to-day variations of the burning velocity as

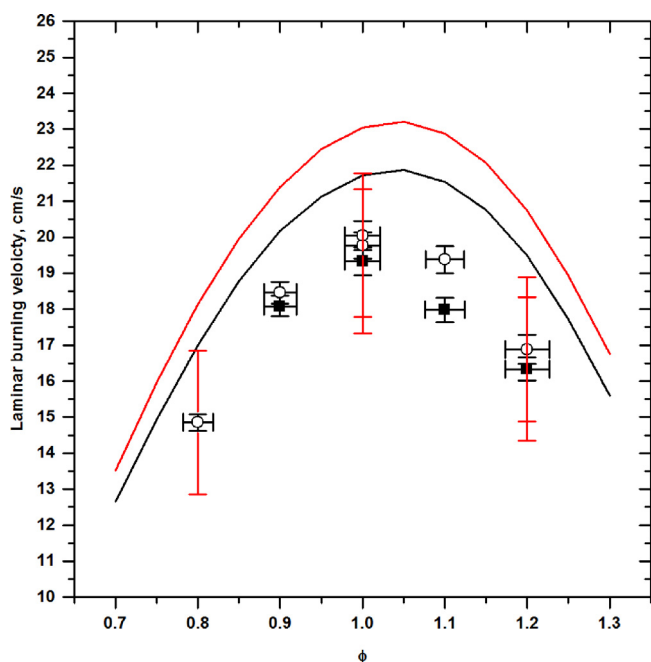


Fig. 12. Laminar burning velocities of acetic acid + air at 338 K. Squares: Day 2; circles: Day 3 experiments. Lines: modeling; red line: Model 1, black line: Model 2. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

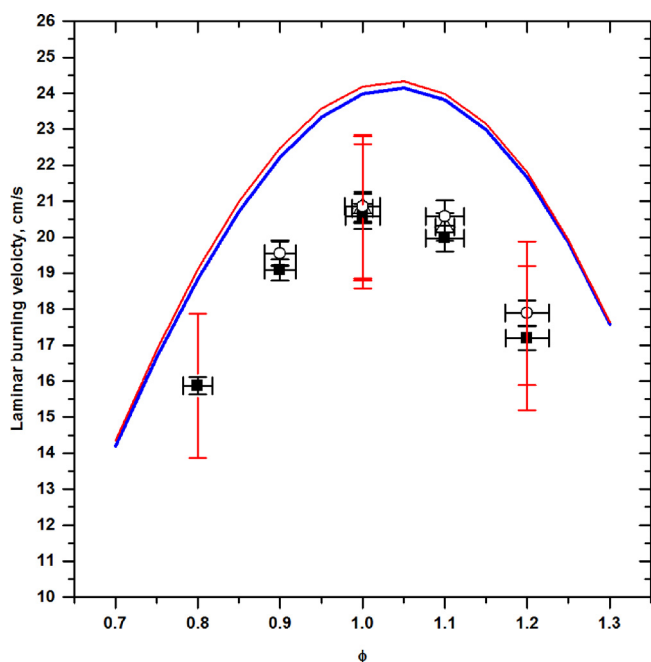


Fig. 13. Laminar burning velocities of acetic acid + air at 348 K. Triangles: Day 1; squares: Day 2; circles: Day 3 experiments. Lines: modeling; red line: Model 1, blue line: Model 3. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

described in Experimental section. It can be seen from Figs. 12–14 that, with exception of two cases, all results are consistent and agree with each other within overlapping uncertainties. On the other hand, the trend of somewhat higher burning velocity on Day 3 is also noticeable. This trend is due to reducing apparent burning velocity of methane flames used for normalization of the measurements, see Fig. S1 in Supplemental material. Lower burning veloc-

ities observed each experimental day are probably caused by reactions of acetic acid with the thermal paste that may lead to modification of the effective surface area of the burner. Formation of patina inside the burner holes may also contribute to the day-to-day variations. However, data scattering of the burning velocities both for methane and acetic acid is always within ± 2 cm/s that indicates the accuracy of the present measurements. The overall uncertainty of the burning velocity measurements includes contributions from the uncertainties of the reference burner “TS”, of the measurements of methane flames on the present burner “E2”, and of the individual measurements of acetic acid flames shown in Figs. 12–14 as smaller error bars. Thus the procedure of S_L recalculation described in Experimental section yields the overall uncertainty of the burning velocity of ± 2 cm/s, also shown in these Figs. and consistent with the data scattering.

Also shown in Figs. 12–14 are calculated burning velocities. Basic mechanism described above and included in the Supplemental material is called Model 1. Model 1 predictions are systematically higher than the measurements with slightly increased deviation in rich flames. The agreement, however, could be considered as satisfactory, given that new experimental data are compared with a new detailed kinetic scheme similarly to the recent study of esters' flames [119]. Models 2, 3 and 4 will be discussed in the following.

To elucidate the role of specific rate constants contributed to the differences between the model and experiments a sensitivity analysis has been performed at 348 K, 1 atm and equivalence ratios 0.7, 1.0 and 1.3. These sensitivity spectra are shown in Fig. 15. None of the acetic acid reactions appears in these spectra. The calculated burning velocities are mostly governed by C1 chemistry typical for all hydrocarbons; the most important are reactions of chain branching $H+O_2=OH+O$, decomposition of formyl radical $HCO+M=H+CO+M$, and oxidation of carbon monoxide $CO+OH=CO_2+H$. Competing with these reactions are recombination of H atoms and oxygen $H+O_2(+M)=HO_2(+M)$, and formation of HO_2 radicals in $HCO+O_2=CO+HO_2$. A discussion regarding the choice of the rate constants for these reactions can be found in [28,29]. These hydrogen and syngas kinetic mechanisms demonstrated good performance in predicting laminar burning velocities at least at conditions close to atmospheric pressure and moderate initial temperatures. Moreover, the present model is in good agreement with the burning velocities of CH_4 + air flames as illustrated in Supplemental material, Fig. S1. One may conclude that the reason of the differences between the model and experiments observed in Figs. 12–14 is in reactions of ketene and ketylenyl radicals.

Reactions $CH_2CO+OH=HCCO+H_2O$ (11) and $HCCO+O_2=CO_2+CO+H$ (35) contribute to additional formation of H atoms from ketene, especially in lean flames. Reaction $CH_2CO+OH=CH_2OH+CO$ (10) is more important in rich flames and also competing since it removes OH radicals producing less reactive CH_2OH . Reaction $CH_2CO+H=CH_3+CO$ (2) effectively converts H atoms into methyl radicals thus decelerating combustion chemistry in all mixtures.

5.2. Low pressure flame

To further analyses performance of the kinetic mechanism, the flame structure of the low-pressure flame of acetic acid [11] has been modeled. Species profiles from a burner stabilized $CH_3CO_2H + O_2 + Ar$ flame at 50 mbar with equivalence ratio 0.9 are presented in Fig. 16. The basic mechanism, Model 1, and three other models described below are also shown. All major products: CO_2 , H_2O , CO , and H_2 are accurately reproduced. The models do not show early reactivity of acetic acid seen experimentally as indicated by the concentration profiles of CH_3CO_2H and OH radicals. The difference between residual calculated and measured concentrations of O_2 in the post-flame zone may be due to uncertainty

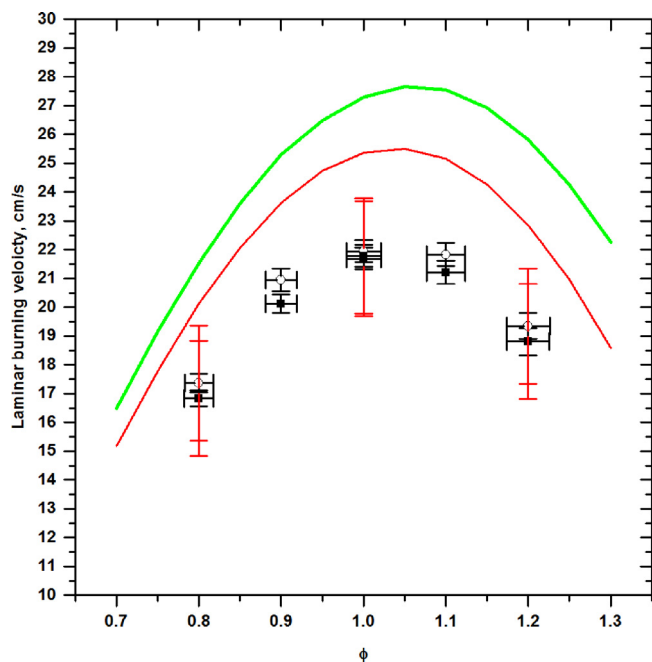


Fig. 14. Laminar burning velocities of acetic acid +air at 358 K. Squares: Day 2; circles: Day 3 experiments. Lines: modeling; red line: Model 1, green line: Model 4. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

in the experimental equivalence ratio. Similar differences were observed with the original model of Leplat and Vandooren [11]. Their model, however, was in much better agreement with the concentration profiles of minor intermediate species: CH_2CO , C_2H_2 , C_2H_4 , CH_3 , and CH_4 , that was achieved by employing non-physical activation energy of reaction $\text{CH}_2\text{CO}_2\text{H} = \text{CH}_2\text{CO} + \text{OH}$ (–13) producing ketene.

Possible modifications of the rate constants aiming at improved agreement with the experiments are outlined below using sensitivity analysis. The sensitivity spectrum for the maximum concentration of CH_2CO in the low-pressure flame shown in Fig. 17 clearly proves that ketene is mostly formed in reaction (–13), and therefore reactions producing $\text{CH}_2\text{CO}_2\text{H}$ radicals are of dominant importance. The rate constants of these reactions between acetic acid and radicals are rather uncertain (see Table 2), and only for the total rate of $\text{CH}_3\text{CO}_2\text{H} + \text{OH}$ are supported by kinetic experiments [110]. Major reaction consuming ketene is $\text{CH}_2\text{CO} + \text{H} = \text{CH}_3 + \text{CO}$ (2), it also contributes significantly to the formation of methyl radicals, as it will be elucidated in the following. Simple reduction of this rate constant against calculated expression [48] will significantly increase concentration of ketene and reduce concentrations of CH_3 and CH_4 thus improving agreement of the present model with the low-pressure flame structure. Ketene will then be consumed mostly in reactions with OH radicals not leading to formation of CH_3 .

To illustrate specific behavior of the kinetic scheme with different rate constants three additional models were tested in addition to the basic mechanism, Model 1. In Model 2 the rate constant of reaction $\text{CH}_2\text{CO} + \text{H} = \text{CH}_3 + \text{CO}$ (2) is increased by a factor of 2. In Model 3 the rate constant of reaction $\text{CH}_3\text{CO}_2\text{H} + \text{OH} = \text{CH}_2\text{CO}_2\text{H} + \text{H}_2\text{O}$ (62) is increased by a factor of 2. In Model 4 the rate constant of reaction $\text{CH}_2\text{CO} + \text{CH}_3 = \text{C}_2\text{H}_5 + \text{CO}$ (19) suggested by Woods and Haynes [33] is adopted. Note that burning velocities calculated using these models are also shown in Figs. 12–14. Directly confirming the outcome of the sensitivity analysis for ketene in low-pressure flame, Models 2 and 4 predict

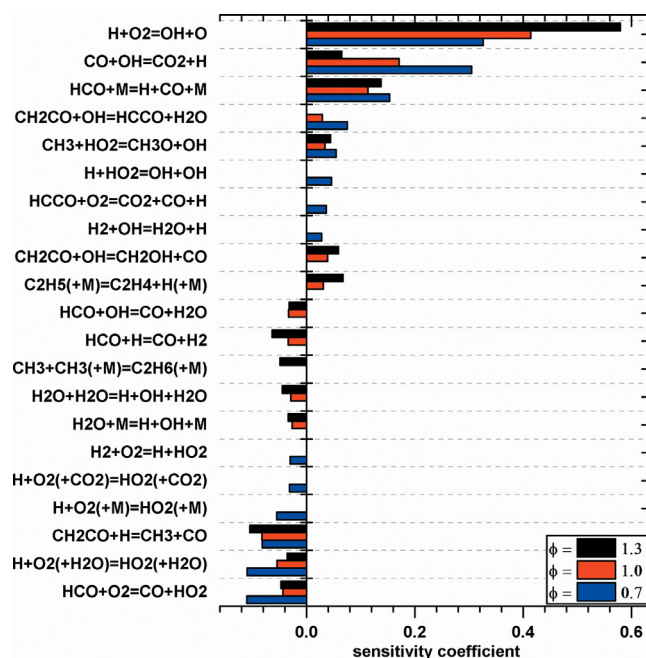


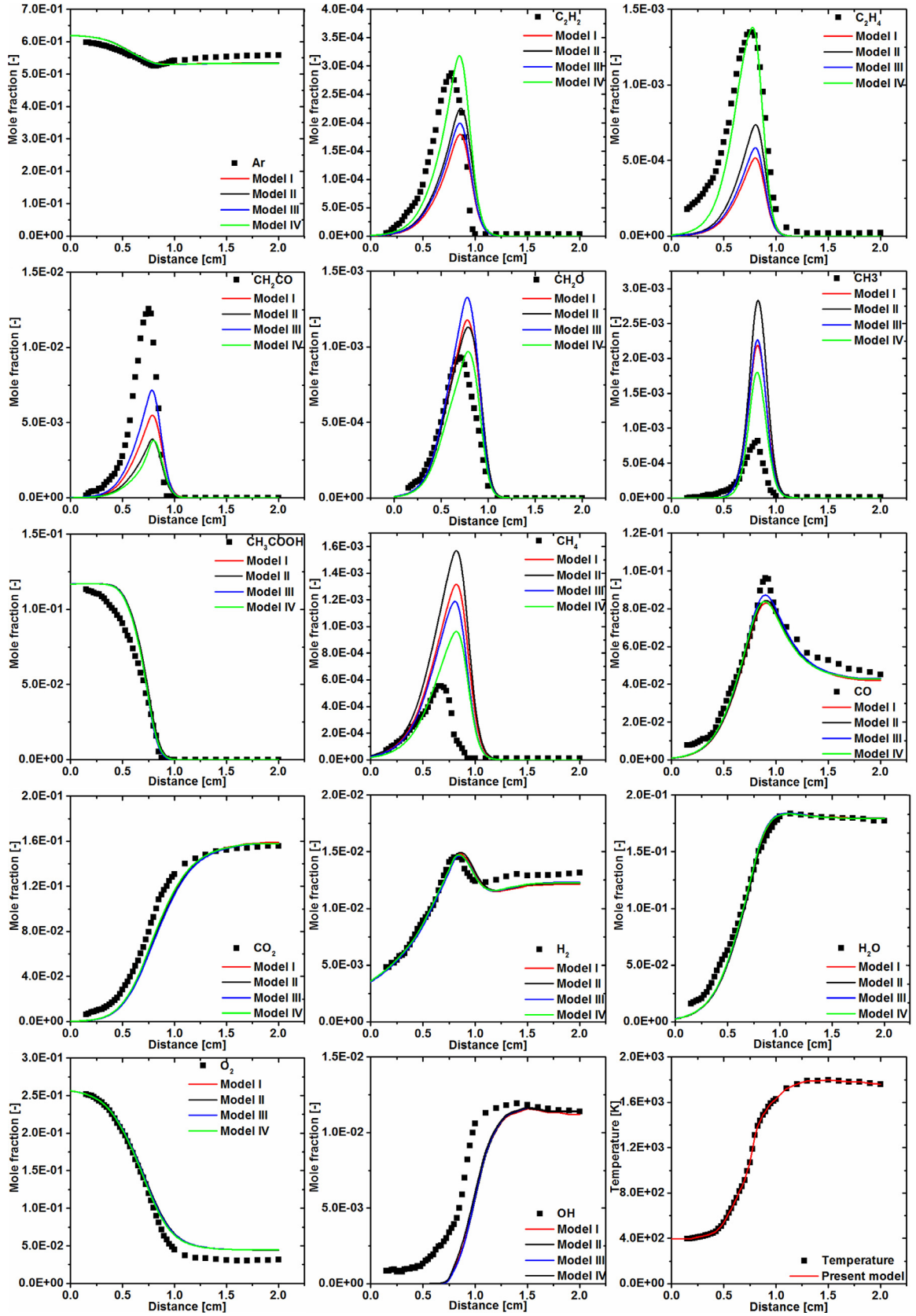
Fig. 15. Normalized sensitivity coefficients for the burning velocity of acetic acid flame at 348 K.

lower concentrations due to increased CH_2CO consumption, while Model 3 shows higher concentration as compared to Model 1 due to increased CH_2CO formation, see Fig. 16.

The sensitivity spectrum for the maximum concentration of CH_3 in the low-pressure flame is shown in Fig. 18. Methyl radicals are major precursors of methane and C2 hydrocarbons. CH_3 is formed in reaction (2) mentioned above and in each decomposition of CH_3CO_2 radicals. Therefore notable decrease of the predicted concentration of CH_3 can be achieved by reducing the rate constant of reaction $\text{CH}_3\text{CO}_2\text{H} + \text{OH} = \text{CH}_3\text{CO}_2 + \text{H}_2\text{O}$ (61).

Methyl radicals may react with acetic acid forming methane; these reactions, however, are not very sensitive for the calculated CH_4 concentration as could be seen from the sensitivity spectrum for methane shown in Fig. 19. Decomposition of acetic acid in reaction $\text{CH}_3\text{CO}_2\text{H} = \text{CH}_4 + \text{CO}_2$ (44) directly contributes to formation of CH_4 ; this reaction does not appear in other sensitivity spectra and can be independently adjusted to improve the model behavior. Model 2 predicts highest concentrations of CH_3 and CH_4 due to enhanced formation in reaction (2). In Model 4 methyl radicals are removed in reaction $\text{CH}_2\text{CO} + \text{CH}_3 = \text{C}_2\text{H}_5 + \text{CO}$ (19) that leads to their efficient conversion into C2 species and consequently to better agreement with the structure of the low-pressure flame, as compared to Model 1.

The sensitivity spectra for the maximum concentration of C_2H_2 and C_2H_4 are shown in Fig. 20 together since relevant reactions are closely connected. Formation of C2 species starts either in recombination of methyl radicals or in their reaction with ketene (19). Therefore, availability of CH_3 and its reactivity in this reaction defines the rate of C2 species formation. If the rate constant of reaction (19) is set very high as suggested by Woods and Haynes [33] and implemented in Model 4, the calculated profiles of C_2H_2 and C_2H_4 closely match experimental concentrations or even exceed them. Figure 20 also shows that the sequence of C2 species conversion is mostly from C_2H_5 to C_2H_4 and C_2H_3 . The balance and maximum concentrations of C_2H_2 and C_2H_4 in particular low-pressure flame is then depends on availability of other flame radicals for H atom abstraction.

Fig. 16. Concentration profiles in a burner stabilized $\text{CH}_3\text{CO}_2\text{H}+\text{O}_2+\text{Ar}$ flame, $\phi = 0.9$.

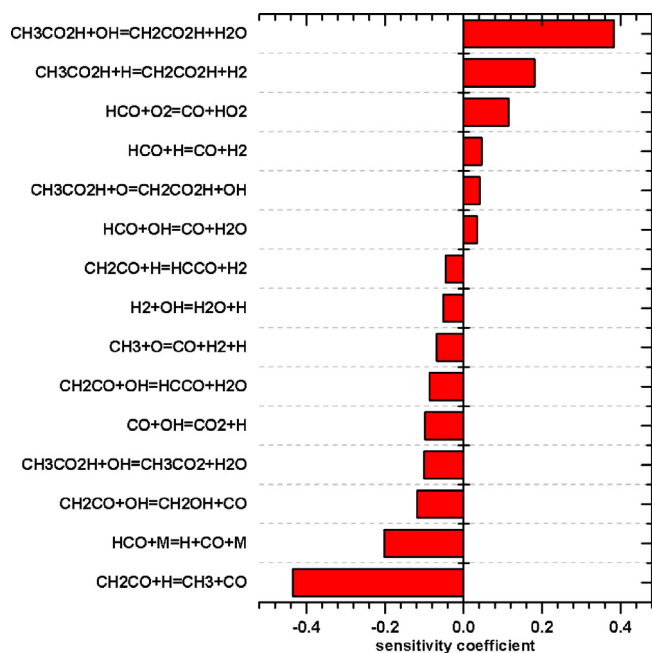


Fig. 17. Normalized sensitivity coefficients for the maximum concentration of CH_2CO .

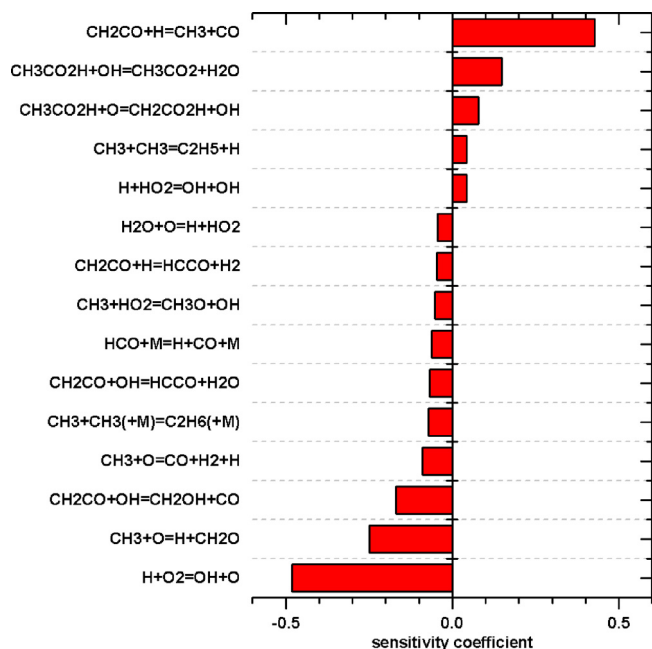


Fig. 18. Normalized sensitivity coefficients for the maximum concentration of CH_3 .

The sensitivity analyses both for the laminar burning velocities and for the species profiles in low-pressure flames clearly indicate the pathways to improve the model behavior by adjusting several rate constants within the evaluated uncertainty factors. These modifications, however, are conflicting as seen from comparison of Models 1–4. Indeed, Model 2 with higher rate constant of reaction $\text{CH}_2\text{CO}+\text{H}=\text{CH}_3+\text{CO}$ (2) is in better agreements with the burning velocities as seen in Fig. 12, yet it predicts too high concentrations of CH_3 and CH_4 . Model 3 is very close to the basic Model 1 in reproducing burning velocities, see Fig. 13; it also shows higher concentrations of ketene in low-pressure flames. However, this is not sufficient to resolve disagreement with the concentrations of minor intermediates: CH_3 , CH_4 , C_2H_2 and C_2H_4 . Model 4

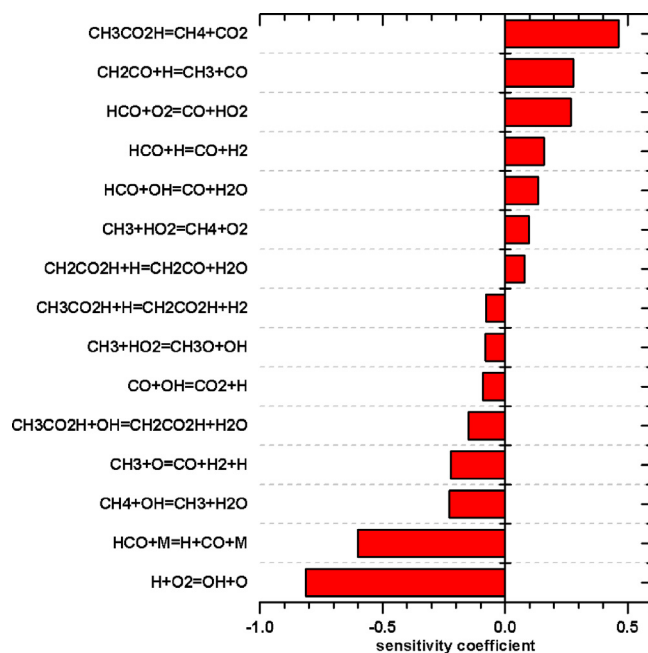


Fig. 19. Normalized sensitivity coefficients for the maximum concentration of CH_4 .

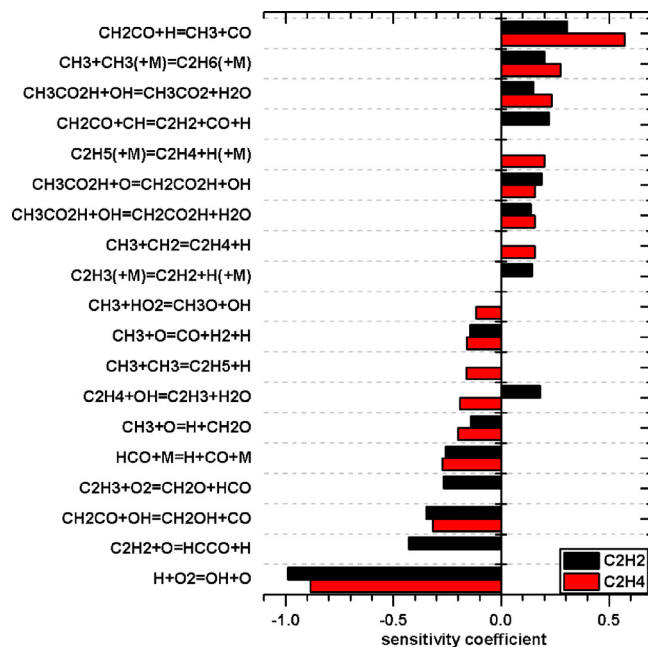


Fig. 20. Normalized sensitivity coefficients for the maximum concentration of C_2H_2 and C_2H_4 .

does a better job in the low-pressure flames, yet it largely overpredicts the burning velocities as shown in Fig. 14. There could be several reasons of this conflicting model behavior. First, reactions of acetic acid decomposition are pressure-dependent [101], and reactions of decomposition of the parent radicals, $\text{CH}_2\text{CO}_2\text{H}$ and CH_3CO_2 could be pressure-dependent too. Thus, model adjustment to reproduce simultaneously atmospheric pressure data and low-pressure species profiles could be misleading until these reactions are properly established and included in the model with pressure-dependent terms. Secondly, reactions of ketene are in the core of acetic acid chemistry. The knowledge of these reactions is still very limited as some key rate constants were simple estimates or were derived by fitting of rather complex kinetic schemes to

experimental data [40,52]. This calls for further experimental studies and analysis, which is the objective of the authors.

6. Conclusions

The laminar burning velocities of acetic acid + air flames at atmospheric pressure and initial gas temperatures of 338 K, 348 K, and 358 K have been determined for the first time. These experiments were accompanied by difficulties caused by the corrosiveness of acetic acid towards the burner material, which forced us to use the burner with poor characteristics. However, the procedure of normalization of the apparent burning velocity values by the well-known burning velocities of methane flames allows to circumvent these problems. Despite of day-to-day variations, data consistency was demonstrated and the uncertainty of the burning velocities was evaluated to be about ± 2 cm/s. No stretch correction was needed as the flames were stabilized on a perforated plate burner at adiabatic conditions using the heat flux method.

A recently updated detailed kinetic mechanism for small oxygenated fuels [17] was extended by reactions of acetic acid and ketene. In total seventy pertinent reactions have been reviewed through analysis and comparison of existing estimations, theoretical calculations and experimental data, if available, to substantiate the choice of particular rate expression with evaluated uncertainty. All theoretical results for H atom abstraction from methyl group for acetic acid and methyl acetate were found consistent showing that close rates could be expected, yet with remaining uncertainty of about a factor of 5. On the other hand, an assumption of the similar reactivity of different radicals in H atom abstraction from acidic group forming CH_3CO_2 and those from alcohol group is apparently erroneous and typically overestimates the rate constants by more than an order of magnitude.

The mechanism was tested comparing with the present measurements and with flame structure of the low-pressure flame of acetic acid [11]. The model over-predicts measured burning velocities by about 3 cm/s. In the low-pressure flames good agreement with the concentration profiles of major products was found, yet several minor intermediates were over- or under-predicted by the model. The sensitivity analyses were performed to elucidate reactions responsible for the differences observed. It was found that the calculated burning velocities are insensitive to the reactions of acetic acid and mostly governed by C1 chemistry typical for all hydrocarbons and by reactions of ketene. The most important reactions are in the sequence of formation of $\text{CH}_2\text{CO}_2\text{H}$ radicals, their decomposition producing ketene, and reactions of CH_2CO with H atoms and methyl radicals. Possible modifications of the rate constants within the evaluated uncertainty factors were discussed, however no attempt to “adjust” the reaction rate expressions was made in the present paper.

Acknowledgments

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.combustflame.2016.05.007](https://doi.org/10.1016/j.combustflame.2016.05.007).

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